

MemEcho: Memory-Aware Lossless KV Prefetching for Native Sparse-Attention LLM Serving

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ABSTRACT

Native sparse-attention models reduce attention computation but retain a KV state that grows with context length. Host offloading can recover accelerator capacity, yet late recall of model-selected KV blocks remains on the critical path. Existing lossless prefetching and hierarchical caches provide strong numerical, query-sequential and, where supported, layer-wise controls. Agent workloads, however, introduce tool calls, memory writes, consolidation, supersession, and task switches that can break this locality. MemEcho asks whether explicit memory-state transitions can predict the exact sparse KV working set before the native indexer consumes it. It uses memory events only for residency and prefetch decisions; the model’s indexer remains authoritative, with authoritative miss handling and identity checks required to preserve the selected KV set and output. This repository first performs a necessity study and admits the mechanism only when matched agent traces expose repeatable memory-transition-induced recall stalls.

1 MECHANISM AND CAUSAL BOUNDARY

A memory event is useful only if it is observable before a later indexer and predicts selected blocks early enough to finish their transfer. Event overlap without lead time is insufficient. The proposed policy changes placement and prefetch timing, never the native indexer’s selection. Each consumed block must match the request, layer, position, version and bytes of the authoritative state; misses wait for exact recall. Cancellation and slot reuse must preserve this identity. No mechanism or runtime correctness proof exists yet.

2 STRONG BASELINES AND NECESSITY

ECHO provides graph-friendly management and lossless prefetching [1]. HiSparse provides an exact hierarchical sparse cache and supports layer-wise prefetch for models with shared selections [2]. Both are mandatory supported controls, alongside full residency and demand-only offload; adjacent layer similarity is not a prerequisite for the proposed event signal.

Hold checkpoint, indexer, graph mode, concurrency, host/device bytes and transfer budgets fixed. Measure event availability, prediction and DMA timing, exposed recall stalls, wasted bytes and request tails. Compare event-conditioned, event-removed and event-shuffled policies; a future-selection oracle supplies headroom only. Preserve the existing gate: at least two representative workloads and at least 20% less synchronous recall or stall than the strongest supported sequential baseline, with identical outputs. The new stronger controls must also be beaten before claiming incremental value. Stop if the signal is late, baseline prefetch hides the stall, or overhead/tail regressions erase benefit. Missing baseline support is reported, not replaced by a weaker port and labeled a win.

3 STATUS

Research contract established; no performance or NPU claim has yet been made.

REFERENCES

- [1] Guangda Liu, Wenhao Chen, Chengwei Li, Zhenyu Ning, Jing Lin, Yiwu Yao, Quan Chen, Shixuan Sun, Jieru Zhao, and Minyi Guo. 2026. ECHO: Efficient KV Cache Offloading with Lossless Prefetching for Serving Native Sparse Attention LLMs. In *USENIX OSDI*. <https://www.usenix.org/conference/osdi26/presentation/liu-guangda>
- [2] Zhiqiang Xie, Zhangheng Huang, Tingwei Huang, Ziyi Xu, Ruiyang Ma, and Christos Kozyrakis. 2026. HiSparse: Scaling Sparse-Attention Decoding with Hierarchical KV Cache Management. <https://arxiv.org/abs/2608.07009>.